

FUNCTIONAL SPECIFICATION DOCUMENT

(FSD)

Medical Diagnosis Center by ServiceNow Service Portal

Project Name	Medical Diagnostic Center by ServiceNow Service Portal
Prepared By	Project Team
Platform	ServiceNow
Document	Functional Specification Document (FSD)
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1. Introduction

This Functional Specification Document defines the functional behavior of the Medical Diagnostic Center implemented on the ServiceNow platform. It converts the demonstrated portal, diagnostic-test catalog, appointment booking, appointment tracking, report access and automation components into a structured functional specification.

The specification is intentionally limited to the demonstrated ServiceNow solution. It does not claim external hospital/laboratory integration, autonomous diagnosis, or an external React/Node/MongoDB application stack.

2. Project Overview

Purpose

Provide a centralized ServiceNow experience through which a patient can discover diagnostic tests, select a test, complete an appointment booking journey, track the appointment and access available diagnostic reports.

Key Features

- Patient-facing Medical Diagnostic Test Center Service Portal.
- Medical Test Catalog for search/browse and diagnostic-test discovery.
- Diagnostic-test information including category, description, duration, availability and price as configured.
- Book Now action from the test catalog.
- Appointment Booking flow with Details → Schedule → Payment → Done stages.
- Structured patient information capture during booking.
- Appointment records stored and tracked through the configured APPOINTMENT TABLE.
- My Appointments page for appointment history, status and details.
- Report records associated with completed diagnostic appointments.
- My Lab Reports page for patient access to available diagnostic report information.
- Business-rule automation for checkout appointment processing, report creation on appointment completion, and rejection-related updates.
- ServiceNow roles and ACLs for controlled access to configured data.

3. Architecture

The functional architecture is organized into the patient-facing Service Portal, diagnostic-test discovery and booking components, ServiceNow data tables, business-rule automation, appointment/report tracking experiences, and security configuration.

Patient	Service Portal	Medical Test Catalog	Appointment Booking	Appointment / Report Data
My Appointments	My Lab Reports	Business Rules	Roles / ACLs	ServiceNow Administration

Figure 1. Functional architecture of the Medical Diagnostic Center.

3.1 User-Facing Layer

The patient-facing layer is the ServiceNow Service Portal. The user enters the Medical Diagnostic Test Center, browses the Medical Test Catalog, selects a diagnostic test and starts the booking journey.

3.2 Application / Automation Layer

The application layer consists of the Appointment Booking experience and configured ServiceNow business rules. The booking flow handles Details, Schedule, Payment and Done stages. Business rules support appointment checkout processing, report creation after

appointment completion and rejection-related updates.

3.3 Data Layer

The solution uses ServiceNow records for the core data. Patient, Diagnostic Test, Appointment and Report information are maintained as structured records with relationships needed for the service lifecycle.

4. Setup Instructions

Prerequisites

- Access to an authorized ServiceNow instance/PDI with application configuration permissions.
- Permission to configure Service Portal pages/widgets, tables, fields, business rules and roles/ACLs.
- Configured patient, diagnostic-test, appointment and report data structures.
- Test users representing patient/requester and authorized operational/administrative roles.
- Access to ServiceNow administrative/debugging views needed to validate business-rule execution.

Configuration Sequence

1. Configure the Medical Diagnostic Test Center Service Portal and required pages/widgets.
2. Configure the Diagnostic Test data structure and Medical Test Catalog experience.
3. Configure the Appointment Booking experience and its Details, Schedule, Payment and Done stages.
4. Configure Patient, Diagnostic Test, Appointment and Report tables/fields required by the workflow.
5. Configure the appointment-related business rules and their conditions/actions.
6. Configure My Appointments and My Lab Reports pages/widgets.
7. Configure required roles and ACLs for patient-facing and administrative access.
8. Create representative test records and validate the complete lifecycle.
9. Perform functional, security and UAT validation.

5. Configuration / Component Structure

Configuration Area	Components	Purpose
Portal	Service Portal pages/widgets	Patient-facing diagnostic-center experience.
Catalog	Medical Test Catalog	Search/browse tests and initiate booking.
Booking	Appointment Booking	Details → Schedule → Payment → Done.
Data Model	Patient / Diagnostic Test / Appointment / Report	Structured storage and lifecycle relationships.
Automation	Business Rules	Checkout processing, report creation and rejection updates.
Tracking	My Appointments / My Lab Reports	Appointment and report access.
Security	Roles / ACLs	Controlled access to configured records.
Administration	ServiceNow configuration	Maintenance, validation and troubleshooting.

6. Functional Requirements

Requirement ID	Requirement	Description
FR-01	Portal Access	The system shall provide a Medical Diagnostic Test Center Service Portal entry point for authorized users.
FR-02	Test Catalog	The system shall provide a searchable/browsable Medical Test Catalog for diagnostic-test discovery.
FR-03	Test Information	The catalog shall display configured diagnostic-test information such as category, description, duration, availability and price where configured.
FR-04	Test Selection	The user shall be able to select a diagnostic test and initiate booking using Book Now.
FR-05	Patient Details	The booking flow shall capture the required patient information configured for the appointment.

Requirement ID	Requirement	Description
FR-06	Schedule	The booking flow shall provide the configured appointment scheduling/slot selection stage.
FR-07	Payment	The booking flow shall provide the configured payment stage before completion.
FR-08	Booking Completion	The booking flow shall provide a Done stage and persist the appointment information.
FR-09	Appointment Tracking	The system shall provide My Appointments for appointment history, status and details.
FR-10	Completion Processing	The system shall support configured appointment completion processing.
FR-11	Report Creation	The configured automation shall support creation/update of report information when an appointment is completed.
FR-12	Lab Reports	The system shall provide My Lab Reports for accessing available diagnostic report information.
FR-13	Rejection Update	The configured rejection-related business rule shall update appointment information according to its condition/action configuration.
FR-14	Security	The system shall use configured ServiceNow roles and ACLs to control access to protected records.
FR-15	Traceability	The system shall maintain relationships between patient, test, appointment and report records sufficient for lifecycle tracking.

7. End-to-End Functional Flow

1. Patient opens the Medical Diagnostic Test Center Service Portal.
2. Patient searches/browses the Medical Test Catalog.
3. Patient selects a diagnostic test and clicks Book Now.
4. Appointment Booking opens at the Details stage.
5. Patient provides the required patient information.
6. Patient selects the required appointment schedule/slot.
7. Patient proceeds through the Payment stage.
8. Booking reaches Done and the appointment record is persisted.
9. Configured business rules process appointment lifecycle events.
10. Patient can review the booking in My Appointments.
11. When the diagnostic appointment is completed, report-creation automation creates/updates report information.
12. Patient can access the available result through My Lab Reports.

8. Authentication & Authorization

Authentication is provided by the configured ServiceNow instance. Authorization is controlled using ServiceNow roles and ACLs. Patient-facing users should only access permitted portal functions and records, while authorized operational and administrative users receive the access required for their responsibilities.

- Patient / requester: browse tests, book an appointment, view permitted appointment information and access permitted reports.
- Operational user: access authorized appointment/report information required for fulfillment.
- Administrator: configure portal components, tables, business rules, roles and ACLs.
- ACL controls: restrict create/read/write/delete access to protected records according to configured role and record context.

9. User Interface

The demonstrated UI consists of the Medical Diagnostic Test Center portal, Medical Test Catalog, Appointment Booking, My Appointments and My Lab Reports.

UI Component	Functional Purpose
Portal Home	Entry point for the diagnostic-center service.
Medical Test Catalog	Search/browse diagnostic tests and Book Now actions.
Appointment Booking	Details → Schedule → Payment → Done.
My Appointments	Appointment history, status and details.
My Lab Reports	Available diagnostic report information.

10. Non-Functional Requirements

Area	Specification
Usability	The portal and booking stages should be clear and understandable to a patient/service user.
Security	Patient, appointment and report information shall be protected by configured ServiceNow access controls.
Reliability	Configured portal components, tables and business rules should execute consistently under normal project test conditions.
Performance	The application should provide responsive portal/form behavior under the tested project conditions; production-scale benchmark values are outside the current scope.
Data Integrity	Patient, diagnostic-test, appointment and report relationships should remain consistent throughout the configured lifecycle.
Maintainability	The solution should remain configurable through native ServiceNow tables, portal components and business rules.
Auditability	ServiceNow record and audit information should support tracing changes and operational activity.
Scalability	The design should allow additional diagnostic tests, services, portal features and integrations to be added later.

11. Testing

Test ID	Test Scenario	Status
FSD-T01	Medical Test Catalog access and display	Pass
FSD-T02	Diagnostic-test search/browse and selection	Pass
FSD-T03	Book Now initiation	Pass
FSD-T04	Patient details capture	Pass
FSD-T05	Appointment scheduling stage	Pass
FSD-T06	Payment stage navigation	Pass
FSD-T07	Booking Done / appointment record creation	Pass
FSD-T08	My Appointments tracking	Pass
FSD-T09	Appointment completion processing	Pass
FSD-T10	Report creation/update automation	Pass
FSD-T11	My Lab Reports access	Pass
FSD-T12	Roles / ACL access validation	Pass
FSD-T13	End-to-end patient workflow	Pass

12. Screenshots / Demonstration

The final demonstration should show the complete patient journey from Service Portal → Medical Test Catalog → Book Now → Patient Details → Schedule → Payment → Done → My Appointments → diagnostic completion → My Lab Reports, followed by administrative verification of the related records and business-rule processing.

13. Known Issues / Current Limitations

Issue / Limitation	Status	Resolution / Note
Production-scale performance benchmark	Not measured	No exact production-scale benchmark is claimed.
External healthcare-system/API integration	Not implemented	Future scope.
External messaging channels	Not implemented	Future scope.
Advanced analytics / BI	Not implemented	Future scope.
Autonomous clinical diagnosis	Not implemented	Outside current scope; clinical decisions remain with authorized professionals.

14. Future Enhancements

- Integration with authorized hospital, EMR/EHR and laboratory systems.
- Automated diagnostic-service fulfilment task creation, assignment and closure.
- Advanced appointment routing, SLA and escalation management.
- Operational dashboards and analytics.
- Approved external email/SMS/WhatsApp notification channels.
- AI-assisted service classification, priority recommendation or anomaly detection with healthcare governance and validation.
- Additional diagnostic-test categories and reusable service components.

Appendix A – Functional Data Mapping

Data Element	Source	Destination	Purpose
Patient	Patient information captured during booking	Patient Table	Identifies the patient associated with the service.
Diagnostic Test	Selected catalog test	Diagnostic Test Table	Identifies the diagnostic service being requested.
Schedule / Slot	Selected booking schedule	Appointment Table	Supports appointment timing and tracking.
Patient + Test	Booking context	Appointment Table	Associates the patient with the selected diagnostic test/appointment.

Data Element	Source	Destination	Purpose
Appointment Status	Workflow/lifecycle state	Appointment Table	Tracks the appointment through its configured lifecycle.
Rejection Reason	Configured appointment update	Appointment Table	Stores rejection-related information where applicable.
Report Number	Report generation process	Report Table	Identifies the diagnostic report.
Report Date	Report generation/update	Report Table	Stores the report date.
Verified By	Report verification	Report Table	Stores the configured verifier relationship.

Appendix B – Current Implementation Boundary

Current scope: ServiceNow Service Portal, Medical Test Catalog, Appointment Booking, Patient/Diagnostic Test/Appointment/Report records, My Appointments, My Lab Reports, configured business rules, roles/ACLs and lifecycle tracking.

Future/not implemented: direct hospital/EMR/EHR/laboratory APIs, external messaging, advanced BI/analytics, trained AI/ML clinical models, autonomous diagnosis/treatment recommendations and production-scale performance benchmarking.

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